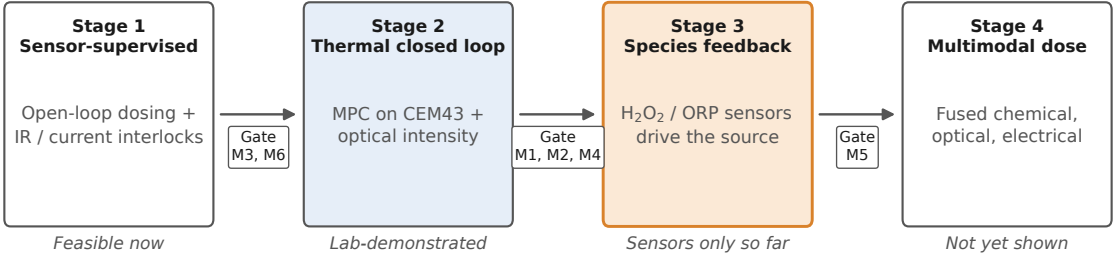
Stage 1
Sensor-supervised
Open-loop dosing +
IR / current interlocks
Gate
M3, M6
Stage 2
Thermal closed loop
MPC on CEM43 +
optical intensity
Gate
M1, M2, M4
Stage 3
Species feedback
H_2O_2 / ORP sensors
drive the source
Gate
M5
Stage 4
Multimodal dose
Fused chemical,
optical, electrical
Feasible now
Lab-demonstrated
Sensors only so far
Not yet shown